

PROGRAM 10 : Demonstrate kubernetes Dashboard and CLI for clustering monitoring

Steps1: Initiate minikube

>minikube start

>cpus 2

>memory 4096

Step2 : enable kubernetes dashboard

>minikube dashboard

Step3: Create a Deployment file

```
GNU nano 7.2
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step4: Create a service file

```
GNU nano 7.2
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
    type: NodePort
```

Step5: apply resource

>kubectl apply -f deployment.yaml

>kubectl apply -f service.yaml

Step6: monitor via CLI

>kubectl get pods

Step7 : check service

>kubectl get svc

Step8 : check pods logs

>kubectl logs <pod-name>

Step9: describe pods

>kubectl describe pod <pod-name>

Step10 : access the application in minikube and viewing the same in the web browser

>minikube service nginx service

```
🔍 Verifying dashboard health ...
🚀 Launching proxy ...
🔍 Verifying proxy health ...
🌐 Opening http://127.0.0.1:39981/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Opening in existing browser session.
^C
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ nano deployment.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ nano service.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/nginx-deployment unchanged
service/nginx-service unchanged
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ kubectl get pods
kubectl get svc
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5j8rb  1/1     Running   2 (132m ago)  2d1h
nginx-deployment-6f9664446b-hzjrm  1/1     Running   2 (132m ago)  2d1h
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP  PORT(S)    AGE
kubernetes                           ClusterIP   10.96.0.1    <none>       443/TCP    14d
nginx-service                        NodePort    10.111.247.118 <none>       80:32375/TCP 165m
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32375

```
🌐 Opening service default/nginx-service in default browser...
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ Opening in existing browser session.
^C
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program10$ kubectl describe pod nginx-deployment-6f9664446b-5j8rb
Name:                               nginx-deployment-6f9664446b-5j8rb
Namespace:                           default
Priority:                             0
Service Account:                       default
Node:                               minikube/192.168.49.2
Start Time:                           Mon, 29 Dec 2025 13:15:52 +0530
Labels:                               app=nginx
                                      pod-template-hash=6f9664446b
Annotations:                           <none>
Status:                               Running
IP:                                   10.244.0.28
IPs:                                  10.244.0.28
Controlled By:                         ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:   docker://8a80a37dda14bb769e6a39cf601243d9699175c9b37468a121d9f16294fcc6a7
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:ca871a86d45a3ec6864dc45f014b11fe626145569ef0e74deaaffc95a3b15b430
    Port:           80/TCP
    Host Port:      0/TCP
```

kubernetes

default

Search

Workloads

Workloads

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Workload Status

Running: 1

Deployments

Running: 2

Pods

Running: 1

Replica Sets

Deployments

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	-	2 / 2	2 days ago

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
nginx-deployment-6f9664446b-5j8tb	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	2	-	-	2 days ago
nginx-deployment-6f9664446b-hzjrm	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	2	-	-	2 days ago

Replica Sets

Name	Images	Labels	Pods	Created
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